

Practical 7 - Competing risks 2

Survival Analysis

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `aaatrial.RData`

R packages required: `survival`, `cmprsk`

```
library(survival)
library(cmprsk)
```

Introduction

In this practical we will use the `aaatrial` data, as in Practical 6. This is a dataset from a randomised trial of screening for abdominal aortic aneurysm (AAA), in which older men were randomised to be invited to be screened for AAA (invited group) or not (control group). The main outcome of interest was deaths relating to AAA (including deaths following AAA repair operations), but men in the study also died from other causes. There are 13,562 people in the data.

If an AAA was found at screening, a repair operation was carried out, essentially removing the risk of future death from AAA. However, the operation itself carries around a 5% risk of mortality. Undetected AAAs may rupture at any time, resulting in either an emergency repair operation (which carries around a 40% risk of mortality) or death. Participants were recruited to the trial between 1997 and 1999 and followed up until the end of 2007.

The key variables are described below.

Variable	Description
id	Unique identifier for each participant
group	Randomisation group: 0=Control, 1=Invited

Variable	Description
dateran	Date of randomisation
aaadeath	AAA-death indicator: 0=Censored or non-AAA-death, 1=AAA-death
allddeath	All-cause mortality: 0=Censored, 1=Death (any cause)
deathtype	0=Censored; 1=non-AAA death; 2=AAA-death
timeout	Date of study exit
dob	Date of birth

This is already an R dataset, and the dates have been formatted for you as dates.

Aims

The aims of this session are:

- Fit and interpret results from cause-specific hazard models.
- Fit and interpret results from subdistribution hazard models.
- Estimate cumulative incidence functions based on cause-specific hazard models and based on a subdistribution hazard model.

Question 1

- Load the data. Create a variable called `time`, which is the follow-up time variable you created in Practical 6, Question 1.
- Fit cause-specific Cox models for AAA-death and non-AAA-death with `group` as the explanatory variable. This can be done using the code below. Note that the `id` option is required when using `coxph` for competing risks.

```
cox.cs<-coxph(Surv(time,as.factor(deathtype))~group,data=aaatrial,id=id)
```

- Write down the form of the hazard models and interpret the cause-specific hazard ratios.
- An equivalent way of fitting these models is by using `coxph` for AAA-deaths and non-AAA-deaths separately, like this:

```
cox.AAA<-coxph(Surv(time,deathtype==2)~group,data=aaatrial)
# cox.AAA<-coxph(Surv(time,aaadeath)~group,data=aaatrial)#same

cox.nonAAA<-coxph(Surv(time,deathtype==1)~group,data=aaatrial)
```

- How is the competing event treated in the fitting of the cause-specific Cox models?

SOLUTION

```
#load the data
load(file="datasets/aaatrial.RData")

#creating time variable
aaatrial$time<-as.numeric(aaatrial$timeout-aaatrial$dateran)/365.25

#Using coxph with deathtype as a factor
cox.cs<-coxph(Surv(time,as.factor(deathtype))~group,data=aaatrial,id=id)
summary(cox.cs)
```

Call:

```
coxph(formula = Surv(time, as.factor(deathtype)) ~ group, data = aaatrial,
      id = id)
```

n= 13562, number of events= 4040

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
group_1:2	-0.02330	0.97697	0.03189	0.03188	-0.731	0.4649
group_1:3	-0.40202	0.66897	0.19921	0.19924	-2.018	0.0436 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
group_1:2	0.977	1.024	0.9178	1.0400
group_1:3	0.669	1.495	0.4527	0.9885

Concordance= 0.504 (se = 0.004)

Likelihood ratio test= 4.69 on 2 df, p=0.1

Wald test = 4.6 on 2 df, p=0.1

Score (logrank) test = 4.66 on 2 df, p=0.1, Robust = 4.66 p=0.1

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

```
#Using coxph for AAA-death and non-AAA-death separately
cox.AAA<-coxph(Surv(time,deathtype==2)~group,data=aaatrial)
summary(cox.AAA)
```

Call:

```
coxph(formula = Surv(time, deathtype == 2) ~ group, data = aaatrial)
```

```
n= 13562, number of events= 105
```

```
      coef exp(coef) se(coef)      z Pr(>|z|)
group -0.4020    0.6690   0.1992 -2.018   0.0436 *
```

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
      exp(coef) exp(-coef) lower .95 upper .95
group    0.669      1.495    0.4527    0.9885
```

```
Concordance= 0.547 (se = 0.024 )
```

```
Likelihood ratio test= 4.16 on 1 df, p=0.04
```

```
Wald test              = 4.07 on 1 df, p=0.04
```

```
Score (logrank) test = 4.13 on 1 df, p=0.04
```

```
# cox.AAA<-coxph(Surv(time,aaadeath)~group,data=aaatrial)#same
```

```
cox.nonAAA<-coxph(Surv(time,deathtype==1)~group,data=aaatrial)
summary(cox.nonAAA)
```

Call:

```
coxph(formula = Surv(time, deathtype == 1) ~ group, data = aaatrial)
```

```
n= 13562, number of events= 3935
```

```
      coef exp(coef) se(coef)      z Pr(>|z|)
group -0.02330    0.97697   0.03189 -0.731    0.465
```

```
      exp(coef) exp(-coef) lower .95 upper .95
group    0.977      1.024    0.9178    1.04
```

```
Concordance= 0.503 (se = 0.004 )
```

```
Likelihood ratio test= 0.53 on 1 df, p=0.5
```

```
Wald test              = 0.53 on 1 df, p=0.5
```

```
Score (logrank) test = 0.53 on 1 df, p=0.5
```

- The form of the cause-specific hazard model for AAA-death (**deathtype=2**) is $h_2(t|X) = h_{0,2}(t)e^{\beta_2 X}$ and that for non-AAA-death (**deathtype=1**) is $h_1(t|X) = h_{0,1}(t)e^{\beta_1 X}$, where

$$h_k(t|X) = \lim_{\delta t \rightarrow 0} \frac{1}{\delta t} \Pr(t \leq T < t + \delta t, D = k | T \geq t), k = 1, 2$$

and where $h_{0,k}(t)$ denotes the cause-specific hazard for event type k ($k = 1, 2$) at time t when $X = 0$.

- For AAA-death, the cause-specific HR for group is 0.67 (95% CI 0.45 to 0.99). This means that at any given time since randomisation, the hazard of AAA-death among individuals invited to screening and who have not yet died from another cause is 0.67 times the hazard for individuals in the control group. The 95% CI excludes the null value of 1, so there is evidence against the null hypothesis that this cause-specific hazard ratio is equal to 1.
- For non-AAA-death, the cause-specific HR for group is 0.98 (95% CI 0.92 to 1.04). This means that at any given time since randomisation, the hazard of non-AAA-death among individuals invited to screening who have not yet died from AAA-death is 0.98 times that among individuals in the control group. The 95% CI includes the null value of 1, so there is no evidence against the null hypothesis that this cause-specific hazard ratio is equal to 1.
- The cause-specific Cox hazard model for AAA-death is fitted using a partial likelihood in which individuals are excluded from the risk set for AAA-death after they have been censored or after they have died from another cause.
- The cause-specific Cox hazard model for non-AAA-death is fitted using a partial likelihood in which individuals are excluded from the risk set for non-AAA-death after they have been censored or after they have had an AAA-death.
- In the fitting of cause-specific hazard models individuals who die from the other cause are treated in the same way as people who are censored. This is fine for estimation of cause-specific hazard ratios.

Question 2

- In this question we will estimate the cumulative incidences of AAA-death and non-AAA-death based on the cause-specific hazard models fitted in question 1. After fitting the cause-specific Cox models as in `cox.cs` (using `deathtype` as a factor variable) we can estimate cumulative incidences as follows:

```
cinc.cs <- survfit(cox.cs, newdata = data.frame(group=c(0,1)))
summary(cinc.cs, times=c(1,5,10))

plot(cinc.cs, lwd=2, lty=c(1,2,1,2), col=c("black", "black", "grey", "grey"))
legend("topleft", c("Control, Non-AAA-death", "Invited, Non-AAA-death",
                   "Control, AAA-death", "Invited, AAA-death"),
      lwd=2, lty=c(1,2,1,2), col=c("black", "black", "grey", "grey"))
```

- What are the estimated cumulative incidences for AAA-death and for non-AAA-death at time 5 for the two treatments.

SOLUTION

```
#estimating cumulative incidences using the cause-specific hazard models
#for group=0,1
cinc.cs <- survfit(cox.cs,newdata = data.frame(group=c(0,1)))

#print cumulative incidences at specified times
summary(cinc.cs, times=c(1,5,10))
```

Call: survfit(formula = cox.cs, newdata = data.frame(group = c(0, 1)))

data 1					
time	n.risk	n.event	Pr((s0))	Pr(1)	Pr(2)
1	13255	300	0.981	0.0177	0.00138
5	11631	1597	0.884	0.1106	0.00516
10	4269	2048	0.747	0.2432	0.00949

data 2					
time	n.risk	n.event	Pr((s0))	Pr(1)	Pr(2)
1	13255	300	0.982	0.0173	0.000927
5	11631	1597	0.888	0.1083	0.003460
10	4269	2048	0.755	0.2388	0.006376

- In the output, Pr(1) is the cumulative incidence for `deathtype=1` (non-AAA death) and Pr(2) is the cumulative incidence for `deathtype=2` (AAA death). Pr(S(0)) is the probability of remaining alive.
- For group 0 (control), the estimated cumulative incidence of non-AAA-death up to 5 years is 11% and the estimated cumulative incidence of AAA-death up to 5 years is 0.5%.
- For group 1 (invited), the estimated cumulative incidence of non-AAA-death up to 5 years is 11% and the estimated cumulative incidence of AAA-death up to 5 years is 0.3%.

Question 3

- Create a new variable that is the participant's age in years at the time of randomisation, using the date of birth (`dob`) and date of randomisation (`dateran`).
- Fit cause-specific Cox models for AAA-death and non-AAA-death with group and age as the explanatory variables.
- Write down the forms of the hazard models and interpret the cause-specific hazard ratios.

SOLUTION

```
#Create variable for age at randomisation
aaatrial$age<-as.numeric(aaatrial$dateran-aaatrial$dob)/365.25

#Cause-specific Cox models including group and age
cox.cs2<-coxph(Surv(time,as.factor(deathtype))~group+age,data=aaatrial,id=id)
summary(cox.cs2)
```

Call:

```
coxph(formula = Surv(time, as.factor(deathtype)) ~ group + age,
      data = aaatrial, id = id)
```

n= 13562, number of events= 4040

	coef	exp(coef)	se(coef)	robust se	z	Pr(> z)
group_1:2	-0.022206	0.978039	0.031887	0.031872	-0.697	0.4860
age_1:2	0.099721	1.104863	0.005554	0.005509	18.101	< 2e-16 ***
group_1:3	-0.399987	0.670329	0.199210	0.199258	-2.007	0.0447 *
age_1:3	0.148551	1.160152	0.034618	0.031759	4.677	2.91e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
group_1:2	0.9780	1.0225	0.9188	1.0411
age_1:2	1.1049	0.9051	1.0930	1.1169
group_1:3	0.6703	1.4918	0.4536	0.9906
age_1:3	1.1602	0.8620	1.0901	1.2347

Concordance= 0.583 (se = 0.005)

Likelihood ratio test= 348.6 on 4 df, p=<2e-16

Wald test = 351.4 on 4 df, p=<2e-16

Score (logrank) test = 351.3 on 4 df, p=<2e-16, Robust = 334.7 p=<2e-16

(Note: the likelihood ratio and score tests assume independence of observations within a cluster, the Wald and robust score tests do not).

- The form of the cause-specific hazard model for AAA-death (deathtype=2) is $h_2(t|X) = h_{0,2}(t)e^{\beta_2 grade + \gamma_2 age}$ and that for non-AAA-death (deathtype=1) is $h_1(t|X) = h_{0,1}(t)e^{\beta_1 grade + \gamma_1 age}$. Here $h_{0,k}(t)$ is the baseline cause-specific hazard at time t for event type k ($k = 1, 2$), referring to a person with group=0 and age=0. The parameter β_k is the cause-specific log hazard ratio for group, and γ_k is the cause-specific log hazard ratio for a 1 year increase in age ($k = 1, 2$).

- AAA-death: The estimated cause-specific HR for group is 0.67 (95% CI 0.45-0.99), and that for age is 1.16 (95% CI 1.09-1.23). The interpretation is that the invited group (`group=1`) are estimated to have a hazard for AAA-death that is 33% lower than that for the uninvited group, conditional on age. The confidence interval excludes 1, so there is evidence of an effect of the intervention of AAA-death hazard, given age. Conditional on group, a 1-year increase in age is associated with a 16% increase in the hazard of AAA-death, and this is highly statistically significant.
- non-AAA-death: The estimated cause-specific HR for group is 0.98 (95% CI 0.92-1.04), and that for age is 1.10 (95% CI 1.09-1.11). The interpretation is that the invited group (`group=1`) is estimated to have a hazard for AAA-death that is 2% lower than that for the uninvited group, conditional on age, but the confidence interval excludes 1, so there is no evidence of an effect of the intervention on the hazard of AAA-death, given age. Conditional on group, a 1-year increase in age is associated with a 10% increase in the hazard of AAA-death, and this is highly statistically significant.

Question 4

Based on the cause-specific hazard models fitted in question 3, obtain the estimated cumulative incidences for AAA-death and for non-AAA-death at time 5 for the following four individuals:

Person	Group	Age
1	0	65
2	1	65
3	0	70
4	1	70

SOLUTION

```
#Cumulative incidences for people aged 65 and 70 at randomisation
#under the two treatments
nd<-data.frame(id=1:4,group=c(0,1,0,1),age=c(65,65,70,70))
cinc.cs2 <- survfit(cox.cs2,newdata = nd)

summary(cinc.cs2, times=c(1,5,10))
```

Call: `survfit(formula = cox.cs2, newdata = nd)`

```
data 1
time n.risk n.event Pr((s0)) Pr(1) Pr(2)
```

1	13255	300	0.988	0.0112	0.00068
5	11631	1597	0.926	0.0718	0.00260
10	4269	2048	0.831	0.1641	0.00496

```

data 2
time n.risk n.event Pr((s0)) Pr(1) Pr(2)
1 13255 300 0.989 0.0110 0.000456
5 11631 1597 0.928 0.0703 0.001747
10 4269 2048 0.836 0.1610 0.003333

```

```

data 3
time n.risk n.event Pr((s0)) Pr(1) Pr(2)
1 13255 300 0.980 0.0184 0.00143
5 11631 1597 0.879 0.1153 0.00534
10 4269 2048 0.735 0.2547 0.00989

```

```

data 4
time n.risk n.event Pr((s0)) Pr(1) Pr(2)
1 13255 300 0.981 0.018 0.000956
5 11631 1597 0.883 0.113 0.003586
10 4269 2048 0.743 0.250 0.006656

```

- The cumulative incidence estimates at time 5 for the four people are shown below.

Person	Group	Age	Cumulative incidence at time 5: AAA-death	Cumulative incidence at time 5: non-AAA-death
1	0	65	0.3%	7%
2	1	65	0.2%	7%
3	0	70	0.5%	12%
4	1	70	0.4%	11%

Question 5

- In this question we will fit Fine and Gray models (i.e. subdistribution hazard models) for AAA-death, using group as the explanatory variable. This can be done using the `crr` function in the `cmprsk` package as follows:

```

crr.AAA <- crr(aaatrial$time, aaatrial$deathtype,
               cov1=aaatrial$group, failcode=2)

```

- Interpret the subdistribution hazard ratios, and think about how the interpretation differs from that of the cause-specific hazard ratios.

SOLUTION

```
crr.AAA <- crr(aaatrial$time, aaatrial$deathtype,
               cov1=aaatrial$group, failcode=2)
summary(crr.AAA)
```

Competing Risks Regression

Call:

```
crr(ftime = aaatrial$time, fstatus = aaatrial$deathtype, cov1 = aaatrial$group,
    failcode = 2)
```

	coef	exp(coef)	se(coef)	z	p-value
aaatrial\$group1	-0.399	0.671	0.199	-2	0.045

	exp(coef)	exp(-coef)	2.5%	97.5%
aaatrial\$group1	0.671	1.49	0.454	0.991

Num. cases = 13562

Pseudo Log-likelihood = -994

Pseudo likelihood ratio test = 4.1 on 1 df,

- The subdistribution HR estimate for group is 0.67 (95% CI 0.45-0.99). The interpretation here is that the group invited to screening have a 33% reduction in hazard of AAA death compared to the control group, *among those at a given time who have not yet died of AAA but who may have died of non-AAA death*. This last part of the previous sentence is the big difference of interpretation compared to the cause-specific hazard model.
- In this example the subdistribution HR for group is very similar to the cause-specific HR. This will be because the incidence of the competing event (non-AAA death) is similar in both groups. If the competing risks differed substantially between groups, we would see a bigger difference between the two types of hazard ratio.

Question 6

- Based on the Fine and Gray model from Question 5, obtain estimates of the cumulative incidences of AAA-death in the two groups using the code below.
- What is the estimated cumulative incidence up to 5 years.

```
cinc.fg.AAA <- predict.crr(crr.AAA, cov1 = c(0,1))
#Note:
#Column 1 is time
#Column 2 is cumulative incidence for AAA in group 0
#Column 2 is cumulative incidence for AAA in group 1
head(cinc.fg.AAA)
```

SOLUTION

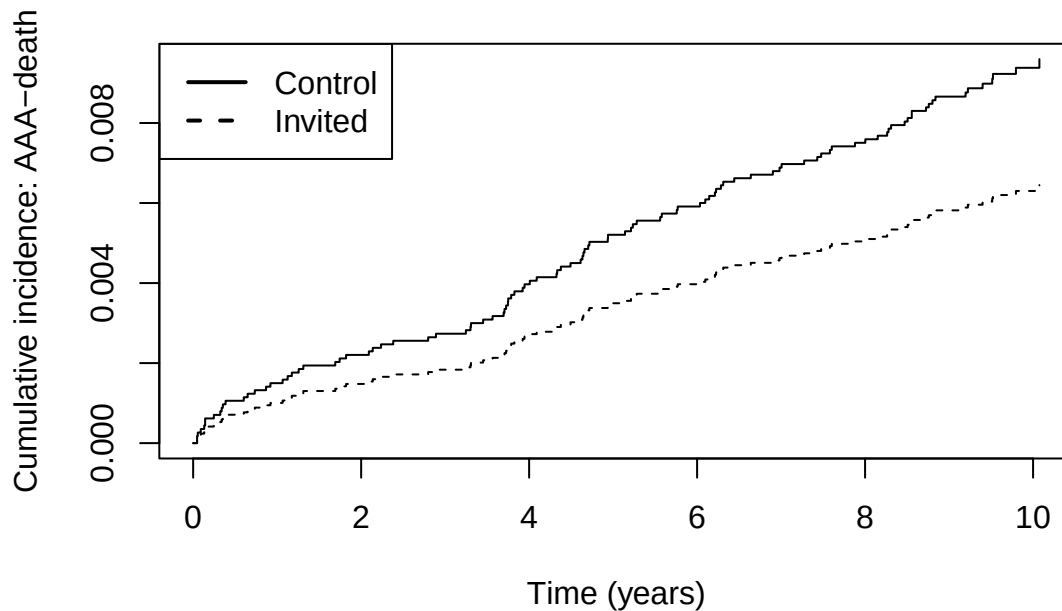
```
cinc.fg.AAA <- predict.crr(crr.AAA, cov1 = c(0,1))
#Note:
#Column 1 is time
#Column 2 is cumulative incidence for AAA in group 0
#Column 2 is cumulative incidence for AAA in group 1
head(cinc.fg.AAA)
```

	[,1]	[,2]	[,3]
[1,]	0.04654346	0.0001763323	0.0001182958
[2,]	0.05475702	0.0002644972	0.0001774455
[3,]	0.09034908	0.0003526700	0.0002366021
[4,]	0.13141684	0.0004408427	0.0002957605
[5,]	0.14236824	0.0006171752	0.0004140736
[6,]	0.24640657	0.0007053428	0.0004732337

```
#Estimates at the time closest to (but before) time 5
cinc.fg.AAA[which.min(abs(cinc.fg.AAA[,1]-5)[cinc.fg.AAA[,1]<5]),]
```

```
[1] 4.939082820 0.005206367 0.003495687
```

```
#plot
plot(cinc.fg.AAA, xlab="Time (years)",
      ylab="Cumulative incidence: AAA-death",
      lwd=2,lty=c(1,2))
legend("topleft",c("Control","Invited"),lty=c(1,2),lwd=2)
```



The estimated cumulative incidence of AAA-death up to 5 years is 0.5% in the control group and 0.3% in the invited group.

Question 7

Using a plot, compare the cumulative incidences of non-AAA-death in the two treatment groups based on three estimation methods: (i) the non-parametric Aalen-Johansen approach, (ii) cause-specific hazard models, (iii) a subdistribution (Fine and Gray) model.

Have a go at making the plot yourself, or see the example code in the solutions.

Explain why you get different estimates using the three approaches.

SOLUTION

```
#(i) Aalen-Johansen estimates (from Practical 6)
cinc.group <- survfit(Surv(time, as.factor(deathtype)) ~ group,
                     data=aaatrial)

time.group0<-cinc.group$time[1:cinc.group$strata[1]]
time.group1<-cinc.group$time[(cinc.group$strata[1]+1):
                             (cinc.group$strata[1]+cinc.group$strata[2])]
cinc.group0<-cinc.group$pstate[1:cinc.group$strata[1],]
cinc.group1<-cinc.group$pstate[(cinc.group$strata[1]+1):
                              (cinc.group$strata[1]+cinc.group$strata[2]),]
```

```

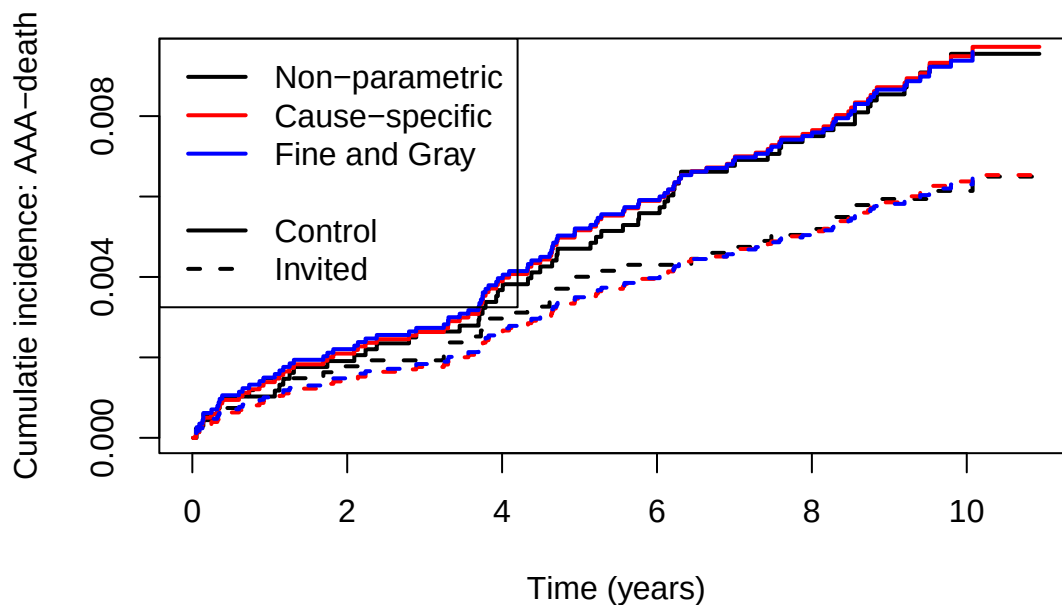
plot(time.group0,cinc.group0[,3],
     type="s",lwd=2,lty=1,col="black",
     xlab="Time (years)",
     ylab="Cumulative incidence: AAA-death")#group 0, AAA-deaths
points(time.group1,cinc.group1[,3],
       type="s",lwd=2,lty=2,col="black")#group 1, AAA-deaths

#(ii) add cause-specific estimates
points(cinc.cs$time, cinc.cs$pstate[,1,3],
       type="s",lwd=2,lty=1,col="red")#group 0, AAA-deaths
points(cinc.cs$time, cinc.cs$pstate[,2,3],
       type="s",lwd=2,lty=2,col="red")#group 1, AAA-deaths

#(iii) add Fine and Gray estimates
points(cinc.fg.AAA[,1], cinc.fg.AAA[,2],
       type="s",lwd=2,lty=1,col="blue")#group 0, AAA-deaths
points(cinc.fg.AAA[,1], cinc.fg.AAA[,3],
       type="s",lwd=2,lty=2,col="blue")#group 1, AAA-deaths

legend("topleft",c("Non-parametric", "Cause-specific", "Fine and Gray",
                  "", "Control", "Invited"),
      col=c("black", "red", "blue", "", "black", "black"),
      lty=c(1,1,1,0,1,2),lwd=2)

```



- In this case the cumulative incidence estimates from the three estimation methods are very similar.
- The non-parametric (Aalen-Johansen) approach does not rely on any modelling assumptions, i.e. it does not assume proportionality of hazards or any particular shape for the hazard.
- The cause-specific hazards approach assumes proportionality of cause-specific hazards between the two groups in the cause-specific models for both AAA-death and non-AAA-death, but does not assume any particular shape for the baseline hazard.
- The Fine and Gray approach assumes proportionality of subdistribution hazards for AAA-death between the two groups, but does not assume any particular shape for the baseline subdistribution hazard.